



2.8

Inverse Functions

Student Notes and Guided Practice

AP Precalculus - Unit 2

| ECHS Mathematics

| Teacher: Mohammad Abu-

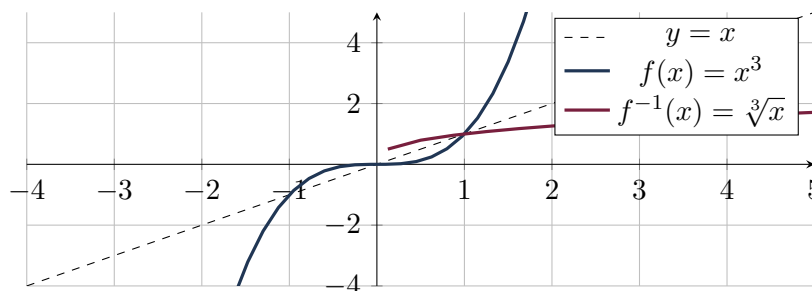
Ghuwaleh

Learning Targets

- Determine whether a function is one-to-one on a stated domain.
- Construct and verify inverse functions.
- Interpret domain/range swapping and contextual restrictions.

Essential Question

When can a function be reversed so each output recovers exactly one input?



Concept Framework

Key Idea

A function has an inverse function exactly when it is one-to-one on its domain.

Key Idea

The horizontal line test checks one-to-one behavior graphically.

Key Idea

Inverse graphs reflect across $y = x$.

Key Idea

Domains and ranges swap; restrictions may be needed.

Formula Map**Formula and Structure**

$$f^{-1}(f(x)) = x$$

Formula and Structure

$$f(f^{-1}(x)) = x$$

Formula and Structure

$$(a, b) \in f \iff (b, a) \in f^{-1}$$

Worked Examples**Worked Example 1**

Problem. Find the inverse of $f(x) = 3x - 7$.

Solution. $f^{-1}(x) = (x + 7)/3$.

Worked Example 2

Problem. Restrict $f(x) = x^2$ so its inverse is a function and find the inverse.

Solution. On $[0, \infty)$, $f^{-1}(x) = \sqrt{x}$.

Worked Example 3

Problem. Why is $1/f(x)$ not inverse notation?

Solution. $1/f(x)$ is the reciprocal of the output, while f^{-1} reverses input and output.

Multiple-Representation Connection**AP Reasoning**

For this topic, always connect at least two representations: algebraic structure, numerical evidence, graphical behavior, or contextual meaning. State restrictions and units before interpreting a result. A correct calculation without a valid domain or contextual interpretation is incomplete AP reasoning.

Common Errors to Avoid

Common Error Check

- Do not discard domain restrictions when rewriting an expression.
- Do not infer local behavior from only one interval-wide average.
- Do not report a numerical answer without units or contextual meaning when quantities are named.
- Do not use a graphing window as proof that a feature does not exist.

Guided Check

1. Find the inverse and state its domain: $f(x) = 5x - 4$.

2. Find the inverse and state its domain: $f(x) = (x - 2)^3 + 1$.

3. Find the inverse and state its domain: $f(x) = \frac{x+1}{x-3}$.

4. Find the inverse and state its domain: $f(x) = x^2 - 4$, $x \geq 0$.

5. Find the inverse and state its domain: $f(x) = 2^x$.

6. Determine whether $f(x) = x^3 - 3x$ is one-to-one on all real numbers.

Lesson Summary

Complete these sentences before leaving the lesson:

1. The most important structure in this topic is _____.
2. A restriction I must remember is _____.
3. One graphical feature I can justify algebraically is _____.